ECE 310 Fall 2026
Lecture 15: Frequency Response

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Learning Objectives

After this lecture, you should be able to:

- Compute the response of LTI systems to sinusoidal input signals.
- Describe and evaluate the response of LTI systems to aperiodic input signals using the DTFT.



Transfer Function and Frequency Response

Previously with the z-transform, we defined the z-transform of an LTI system's impulse response as the *transfer function* of that system.

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Now with the DTFT, we define the DTFT of an LTI system's impulse response as the *frequency response* of that system.

$$\text{Impulse response } h[n] \overset{\mathcal{F}}{\longleftrightarrow} H_d(\omega) \text{ Frequency response.}$$

Thus, in general for aperiodic inputs: $y[n] = x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} Y_d(\omega) = X_d(\omega) H_d(\omega)$



Magnitude and Phase Response

The frequency response of an LTI system $H_d(\omega)$ is a complex-valued function at each frequency $\omega \in [-\pi, \pi]$. Therefore, the frequency response has both a magnitude and a phase at each frequency.

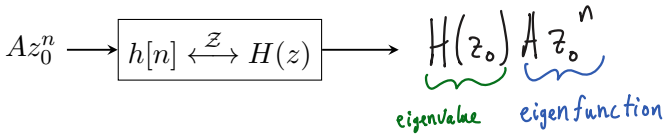
- Magnitude Response $|H_d(\omega)| = \left|\sqrt{H_d^*(\omega) H_d(\omega)}\right| \geq 0$
- Phase Response $\angle H_d(\omega) = \begin{cases} \tan^{-1}\left(\frac{b}{a}\right), & a \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) + \pi, & a < 0,\ b \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) - \pi, & a < 0,\ b < 0 \end{cases}$

  $a = \mathrm{Re}\{H_d(\omega)\}$
  $b = \mathrm{Im}\{H_d(\omega)\}$
- In general, $H_d(\omega) = |H_d(\omega)| e^{j \angle H_d(\omega)}$

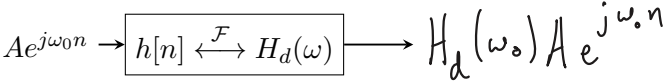


LTI System Response to Periodic Exponential Signals

With the z-transform, we previously showed that for any $z_0 \in \mathbb{C}$



We know that the DTFT is the z-transform evaluated along the unit circle, i.e. for each z_0 on the unit circle $z_0 = e^{j\omega_0}$, thus



Example: LTI System Response to Periodic Exponential Signals

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Consider an LTI system with the following magnitude and phase response.

$$|H_d(\omega)| = |\omega|, \ \angle H_d(\omega) = -\frac{\omega}{2}, \ -\pi \le \omega \le \pi.$$

(a) Determine the frequency response $H_d(\omega)$.

$H_d(\omega) = |H_d(\omega)| e^{j\angle H_d(\omega)} = |\omega| e^{-j\frac{\omega}{2}}$

(b) Determine the system response $y[n]$ when $x[n] = 2e^{j\frac{\pi}{3}n} + e^{j(\frac{\pi}{2}n + \frac{\pi}{4})}$.

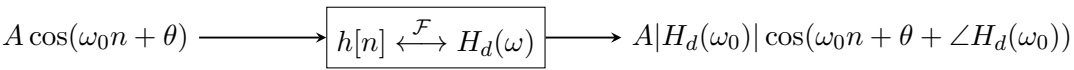
$y[n] = H_d(\frac{\pi}{3}) \cdot 2e^{j\frac{\pi}{3}n} + H_d(\frac{\pi}{2}) \cdot e^{j(\frac{\pi}{2}n + \frac{\pi}{4})}$

$\rightarrow H_d(\frac{\pi}{3}) = \frac{\pi}{3} e^{-j\frac{\pi}{6}}$

$\rightarrow H_d(\frac{\pi}{2}) = \frac{\pi}{2} e^{-j\frac{\pi}{4}}$

$y[n] = \frac{2\pi}{3} e^{j(\frac{\pi}{3}n - \frac{\pi}{6})} + \frac{\pi}{2} e^{j\frac{\pi}{2}n}$

Is the following then also true?



if and only if
h[n] is real-valued.

LTI System Response to Sinusoidal Signals

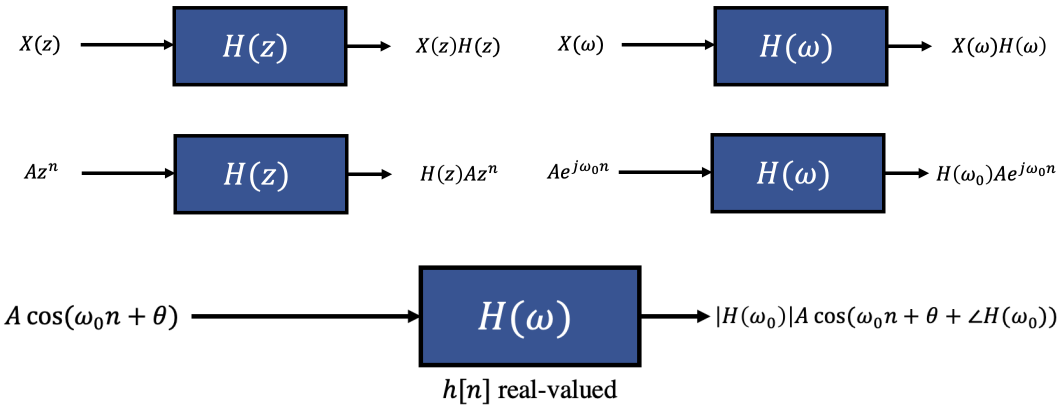
Why does $h[n]$ need to be real-valued?

$$
\begin{aligned}
y[n] &= x[n] * h[n] \\
x[n] &= \cos(\omega_0 n + \theta) = \frac{1}{2}\left(e^{j(\omega_0 n+\theta)} + e^{-j(\omega_0 n+\theta)}\right) \\
y[n] &= \frac{1}{2}\left(H_d(\omega_0)e^{j(\omega_0 n+\theta)} + H_d(-\omega_0)e^{-j(\omega_0 n+\theta)}\right) \\
&= \frac{1}{2}\left(|H_d(\omega_0)|e^{j(\omega_0 n+\theta+\angle H_d(\omega_0))} + |H_d(-\omega_0)|e^{-j(\omega_0 n+\theta-\angle H_d(-\omega_0))}\right) \\
h[n] \text{ real-valued} \implies &= \frac{1}{2}\left(|H_d(\omega_0)|e^{j(\omega_0 n+\theta+\angle H_d(\omega_0))} + |H_d(\omega_0)|e^{-j(\omega_0 n+\theta+\angle H_d(\omega_0))}\right) \\
&= |H_d(\omega)|\cos\left(\omega_0 n + \theta + \angle H_d(\omega_0)\right)
\end{aligned}
$$

Hermitian Symmetry



LTI System Response to Periodic Signals Summary



Concept Check: LTI System Response to Periodic Signals

Consider an LTI system with the following magnitude and phase response.

$$|H_d(\omega)| = \cos^2(\omega),\ \angle H_d(\omega) = \sin(\omega),\ -\pi \le \omega \le \pi.$$

(a) Is this system real-valued?

→ $|H_d(\omega)|$ is even-symmetric ↘
→ $\angle H_d(\omega)$ is odd-symmetric ↗ $h[n]$ is real-valued

Hermitian symmetry
$H_d^*(\omega) = H_d(-\omega)$
or
$|H_d(\omega)| = |H_d(-\omega)|$ and $\angle H_d(\omega) = -\angle H_d(-\omega)$

(b) Determine the system response $y[n]$ for $x[n] = \sin\left(\frac{\pi}{3}n\right) + \cos\left(\frac{\pi}{2}n + \frac{3\pi}{10}\right) + (-1)^n$.

$y[n] = \frac{1}{4}\sin\left(\frac{\pi}{3}n + \frac{\sqrt{3}}{2}\right) + (-1)^n$

→ $|H_d(\frac{\pi}{3})| = \frac{1}{4}$, $\angle H_d(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$

→ $|H_d(\frac{\pi}{2})| = 0$, $\angle H_d(\frac{\pi}{2}) = 0$

→ $(-1)^n = e^{j\pi n}$ → $H_d(\pi) = |H_d(\pi)| e^{j\angle H_d(\pi)}$
$= 1 \cdot e^{j0}$
$= 1$

Example: LTI System Response to Periodic Signals

The system response $y[n]$ to input $x[n]$ is given below for a real-valued LTI system.

$$x[n] = 5 + \cos\left(\frac{\pi}{4}n - \frac{\pi}{3}\right) + \sin\left(-\frac{2\pi}{3}n\right)$$

$$y[n] = -10 + \cos\left(-\frac{2\pi}{3}n\right)$$

Determine the system response to the input $v[n] = -1 + \sin\left(-\frac{\pi}{4}n\right) + 2\cos\left(\frac{2\pi}{3}n\right)$

- $H_d(0) = -2$
- $|H_d(\frac{\pi}{4})| = 0 \rightarrow |H_d(-\frac{\pi}{4})| = 0$ ← by Hermitian symm
- $|H_d(-\frac{2\pi}{3})| = 1$, $\angle H_d(-\frac{2\pi}{3}) = \frac{\pi}{2}$

↳ $|H(\frac{2\pi}{3})| = 1$, $\angle H(\frac{2\pi}{3}) = -\frac{\pi}{2}$

$\rightarrow y[n] = (-2)(-1) + (0)(\sin(-\frac{\pi}{4}n)) + (1)(2\cos(\frac{2\pi}{3}n - \frac{\pi}{2}))$

$y[n] = 2 + 2\cos\left(\frac{2\pi}{3}n - \frac{\pi}{2}\right)$

$\sin\left(\frac{2\pi}{3}\right)$